

Steal IQ: Catalog

Every folder, data file, result and figure in the repository: what it is and which code uses it

1. The repository

The folders are a research funnel: each stage reads only what the stage before it wrote. Results and figures are rebuilt by the code and are not committed; this catalog lists them as they stand on disk, and is rebuilt with them (python3 2-Data-Analysis/build_catalog.py).

Folder	What it holds	Start with
1-Data-Ingestion	Scrapes Baseball Savant and the MLB StatsAPI into committed raw tables and one meta table per season; times the pitcher's delivery from broadcast video (vision/).	ingest.py, vision/delivery.py, data/meta/
2-Data-Analysis	The models and the paper: core.py (shared paths, loaders and models), paper.py (the paper's mixed models), research.py (the archived report's studies), stealq.py (command line), and the build_*.py scripts that write the paper, this catalog and the analysis notebook.	stealiq.py, StealIQ_Paper.ledger.md
3-Presentations	The SABR seminar prototype; the deck itself embeds broadcast video and stays on this machine.	Saber_Prototype.pdf
4-Archive	Superseded code, reports, video-timing logs and outputs; on this machine only (gitignored).	README.md
5-Live-Webapp	Builds the live site's calculator and leaderboard and writes them into docs/index.html.	webapp.py, README.md
docs	The live site, served by GitHub Pages from this folder.	index.html

2. Data

File	Key	Rows × cols	Used by	What it is
1-Data-Ingestion/data/raw: committed source tables				
DF_v7_SSSI.csv	runner_id, season	673 × 66	ingest.py, research.py, webapp.py	Legacy v7 season features (runner skill): an input to Raw_Season and the engines.
DF_v7_xSB_Outcome.csv	runner_id, season	1,300 × 14	ingest.py	Legacy v7 expected-SB outcomes: an input to Raw_Season.
Raw_Attempt_Context.csv	play_id	13,422 × 9	delivery.py, ingest.py, research.py	MLB-feed context per attempt: hand, batter side, pitch code, outs, inning. balls/strikes are post-pitch and score_diff end-of-PA: never inputs.
Raw_Attempts.csv	play_id	13,430 × 16	delivery.py, ingest.py	Every tracked running event 2023+ from Savant's drawer (SB/CS + PK/BK/FB): runner, pitcher, catcher, base, result, lead at first move, ground gained to release. The canonical attempt order.
Raw_Opportunities.csv.gz	play_id	617,453 × 16	build_paper.py, ingest.py, research.py	Every pitch with a runner on 1st and 2nd empty: the decision-model denominator.
Raw_Season.csv	runner_id, season	779 × 77	build_paper.py, core.py, ingest.py, research.py, webapp.py	Runner-season table: sprint speed and running splits, SB, CS, attempts, team (v7 features + live overlay).
Raw_Statcast_Pitches.csv.gz	play_id	12,433 × 120	ingest.py	The Statcast row (all columns) of every attempt pitch, matched via the MLB feed's (at_bat_number, pitch_number).

File	Key	Rows × cols	Used by	What it is
league_sb_rates.csv	season	12 × 7	build_paper.py, ingest.py, research.py	MLB regular-season SB, CS, games, success % and attempts per game (StatsAPI, all 30 teams).
people.csv	player_id	879 × 2	ingest.py, research.py	MLB names for every runner and batter id (StatsAPI).
poptime.csv	catcher_id, season	952 × 8	core.py, ingest.py, research.py, webapp.py	Savant catcher pop time, arm strength and exchange.
sprint_speed.csv	runner_id, season	6,671 × 3	analysis.ipynb, build_notebook.py, build_paper.py, core.py, ingest.py, paper.py, research.py, webapp.py	Statcast sprint speed for every runner.
statcast_csv_docs.csv	field	115 × 4	ingest.py	Savant's own definition of each Statcast CSV field (baseballsavant.mlb.com/csv-docs).
team_map.csv	runner_id, season	779 × 3	ingest.py	Runner's team by season (site display).
1-Data-Ingestion/data/meta: the per-season tables				
data_dictionary.csv	column	171 × 5	analysis.ipynb, build_notebook.py, ingest.py	Every meta-table column: block, source, description, % non-null; dropped columns listed.
meta_2023.csv	play_id + runner_id	3,290 × 163	analysis.ipynb, core.py	THE 2023 TABLE: one row per tracked running event, every Statcast field, MLB-feed context, season pop time and sprint speed, CV delivery time, video links, qa_flags.
meta_2024.csv	play_id + runner_id	3,359 × 163	analysis.ipynb, core.py	THE 2024 TABLE: one row per tracked running event, every Statcast field, MLB-feed context, season pop time and sprint speed, CV delivery time, video links, qa_flags.
meta_2025.csv	play_id + runner_id	3,220 × 163	analysis.ipynb, core.py	THE 2025 TABLE: one row per tracked running event, every Statcast field, MLB-feed context, season pop time and sprint speed, CV delivery time, video links, qa_flags.
meta_2026.csv	play_id + runner_id	3,561 × 163	analysis.ipynb, core.py	THE 2026 TABLE: one row per tracked running event, every Statcast field, MLB-feed context, season pop time and sprint speed, CV delivery time, video links, qa_flags.
1-Data-Ingestion/data/delivery: video delivery times				
delivery_2023.csv	play_id	3,065 × 19	build_paper.py, core.py, ingest.py, research.py	CV delivery time for every 2023 SB/CS attempt: QA verdict, frames, delivery_s, confidence.
delivery_2024.csv	play_id	3,114 × 19	build_paper.py, core.py, ingest.py, research.py	CV delivery time for every 2024 SB/CS attempt: QA verdict, frames, delivery_s, confidence.
delivery_2025.csv	play_id	2,997 × 19	build_paper.py, core.py, ingest.py, research.py	CV delivery time for every 2025 SB/CS attempt: QA verdict, frames, delivery_s, confidence.
delivery_2026.csv	play_id	3,266 × 19	build_paper.py, core.py, ingest.py, research.py	CV delivery time for every 2026 SB/CS attempt: QA verdict, frames, delivery_s, confidence.
delivery_gold.csv	play_id	50 × 16	analysis.ipynb, build_notebook.py, build_paper.py, delivery.py, research.py	The CV scored on the 50 hand-labelled gold clips (accuracy check).

3. Results

File	Rows × cols	What it is
2-Data-Analysis/results/1-Paper-GLMM		
DF_paper_anatomy.csv	1 × 8	The median steal of second: lead, ground gained, sprint speed, delivery, pitch flight, pop time (paper, Fig 1).
DF_paper_breakeven.csv	6 × 7	Break-even success rate for steals of second and third by outs (paper, Fig 2, Table A2).
DF_paper_calibration_all.csv	71 × 5	2026 calibration bins of every model (paper, Fig 4B).
DF_paper_decision_calibration.csv	62 × 6	2026 calibration bins of the decision GLMM by base.
DF_paper_decision_fixed.csv	42 × 16	Decision GLMM fixed effects by base (paper, Fig 5, Table B2); rows with model = evaluation are the superseded decision-plus-ground-gained model.
DF_paper_decision_forward.csv	8 × 13	Decision GLMM forecast test by base against simpler pre-pitch models (calibration quoted in paper §4.3).
DF_paper_decision_players.csv	5,782 × 9	Every player's shrunk effect in the decision model (paper, Table 5).
DF_paper_decision_value.csv	14 × 10	2026 runs per attempt by the decision model's margin over break-even, and go / hold totals (paper, Fig 10).
DF_paper_decision_variance.csv	12 × 11	Decision GLMM random-intercept SDs with likelihood-ratio intervals (paper, Fig 9A, Table B3).
DF_paper_forward_all.csv	8 × 17	THE HEADLINE: every model fit on 2023-25 and scored on the same 2026 attempts: AUROC (clustered CI, by base), log-loss, Brier, calibration, paired difference from v16 (paper, Table 2, Fig 4).
DF_paper_glmm_check.csv	13 × 6	GLMM recovery on 20 simulated data sets and the statsmodels cross-check (paper, Table C1).
DF_paper_greenlight.csv	256 × 20	2027 green-light list: each recent runner's predicted success by outs, average and tough battery (paper, Fig 11, Table 4).
DF_paper_ground.csv	18 × 6	Ground needed by release to clear each break-even rate, by base, outs and catcher, and the share of attempts reaching it (paper, Fig 6).
DF_paper_influence.csv	42 × 10	Each input's log-odds weight per SD (indicators 0 to 1), decision and success models, by base (paper, Fig 5).
DF_paper_pop_choice.csv	2 × 5	Steals of third: pop time to third vs to second in the decision model, same attempts (paper, Table C3).
DF_paper_re24.csv	96 × 6	Runs to the end of the half-inning by season and base-out state: the run-expectancy table (paper, Appendix A).
DF_paper_roc.csv	1,407 × 3	ROC curves of every model on 2026 (paper, Fig 4A).
DF_paper_runner_speed.csv	1 × 9	Runner-season speed vs success: correlation with and without Josh Naylor, binomial noise, reliability, true SD, corrected r (paper, Fig 8).
DF_paper_speed_2b.csv	4 × 12	Sprint speed and steals of second by season: slope, p, swing, AUROC, runner-level r (paper, Fig 7).
DF_paper_speed_bins.csv	11 × 15	Steals of second by sprint-speed bin: stolen bases and caught stealing by season, success, runner-seasons, attempts per runner-season (paper, Table 3).
DF_paper_success_cv.csv	6 × 7	Choosing the success model: three specifications, five-fold CV on 2023-25 only, per base (paper, Table C2).
DF_paper_success_fixed.csv	28 × 9	Success GLMM fixed effects by base, 2023-26, with 10th/90th percentiles (paper, Table B1).
DF_paper_success_players.csv	2,890 × 8	Every player's shrunk effect in the success model: skill net of the jump and the pitch.
DF_paper_success_variance.csv	6 × 9	Success GLMM random-intercept SDs with likelihood-ratio intervals (paper, Fig 9B, Table B3).
2-Data-Analysis/results/2-v16-Logistic		
DF_v16_base.csv	1 × 23	Base-stolen test (technical report §3.3).
DF_v16_base_slopes.csv	4 × 19	Separate slopes for 2nd vs 3rd (rejected).

File	Rows × cols	What it is
DF_v16_engine_leak_fix.csv	27 × 6	One-off record: v12/v13 before vs after the score-leak fix.
DF_v16_final_trials.csv	200 × 11	Every trial of the final Optuna study.
DF_v16_forward.csv	4 × 11	v16 forward test: train 2023–25, test 2026.
DF_v16_leakage_audit.csv	11 × 4	When each candidate input is known (technical report §2.6).
DF_v16_meta.json	—	v16 run summary: final params, importances, MLE coefficients, delivery beta.
DF_v16_models.csv	6 × 20	v16 HEADLINE: shipped → + pitch type → + base, tuned, isotonic (technical report §3.2).
DF_v16_nested_choices.csv	5 × 9	Configuration chosen in each outer fold.
DF_v16_pop_prior.csv	2 × 6	Same- vs prior-season pop time.
DF_v16_recal.csv	15 × 14	Drift recalibration methods (technical report §2.5).
DF_v16_v12_score_audit.csv	3 × 7	One-off audit: v12 with no / end-of-PA / clean score.
DF_v16_v13_score_audit.csv	3 × 9	One-off audit: v13 with no / end-of-PA / clean score.
DF_v16_weights.csv	7 × 15	v16 input weights on one scale (technical report §3.4).
2-Data-Analysis/results/3-Engines-v12-v13		
DF_v12_AUC.csv	9 × 5	v12 success engine: AUROC by feature block.
DF_v12_Calibration.csv	20 × 6	v12 calibration by decile.
DF_v12_Validation.csv	3 × 5	v12 under random / grouped / forward splits.
DF_v13_Attempt.csv	6 × 7	v13 decision engine: AUROC/AUPRC by block and split.
DF_v13_Calibration.csv	10 × 5	v13 forward calibration.
DF_v13_SelectionEffect.csv	5 × 9	Attempt rate by sprint-speed quintile.
DF_v13_SpeedPercentiles.csv	9 × 2	Sprint-speed percentiles in the opportunity population.
2-Data-Analysis/results/4-Studies		
DF_add_algo.csv	8 × 9	Logistic vs splines vs XGBoost vs LightGBM (technical report §2.2).
DF_add_context_prepitch.csv	21 × 16	Pre-pitch model with game context.
DF_add_delivery_beta.csv	6 × 25	Delivery-time coefficient five ways (technical report §6.2).
DF_add_jump_metric.csv	8,557 × 14	Per-attempt jump speed (technical report §6.4).
DF_add_jump_runners.csv	14 × 12	Runners' ground split into jump-speed and pitcher-time parts.
DF_add_jump_tests.csv	37 × 7	Jump-speed tests: decomposition, reliability, out-of-sample value.
DF_add_meta.json	—	Verdict, reproduced coefficients, delivery summary.
DF_add_needle.csv	8 × 31	Six pre-pitch inputs on one scale (technical report §6.3).
DF_add_pitchtype.csv	3 × 15	Safe rate and observed – expected by pitch type (technical report §3.3).
DF_add_pitchtype_state.csv	36 × 11	Pitch type × game state cells and interaction tests.
DF_add_prepitch.csv	12 × 11	2023 delivery profile in a pre-pitch model (technical report §6.3).
DF_add_prior_delivery.csv	33 × 8	Earlier delivery as a pre-pitch input (technical report §6.5).
DF_add_speed_suppression.csv	31 × 24	Speed coefficient M0–M3, mediation, random-effect fits, speed bands (technical report §4).
2-Data-Analysis/results/5-Public-Baselines		
DF_cmp_baselines.csv	11 × 12	Public baselines vs this model (technical report §5.2).
DF_cmp_ladder.csv	7 × 6	What each input adds, and why (technical report §3.1).
DF_cmp_runner_level.csv	4 × 6	Runner-season check of each model (technical report §5.4).
DF_cmp_scenarios.csv	4 × 9	Four real runner-seasons under each model.
DF_cmp_seasons.csv	12 × 11	League success and attempts per game by season (technical report §1.4).
DF_cmp_speed_bins.csv	51 × 10	Per season × 0.5 ft/s speed bin: attempts, SB, CS, runners, share safe with Wilson CI (technical report §4.4).

File	Rows × cols	What it is
DF_cmp_speed_by_season.csv	5 × 25	Sprint speed vs success by season and pooled: speed-only logit (clustered p), 10th→90th swing, AUROC, the same for ground gained, runner-level r (technical report §4.4).
5-Live-Webapp/results		
DF_calculator.csv	29 × 9	THE CALCULATOR: the success GLMM's fixed part per base, term by term as the page runs it, and its 2026 forward-test AUROC.
DF_perattempt_AUC.csv	4 × 2	Per-attempt XGBoost AUROC ladder.
DF_perattempt_Importance.csv	8 × 2	Per-attempt XGBoost feature importance.
DF_success_model.csv	6 × 3	v15 four-input logistic on 2023-26 (intercept, 4 coefficients, CV AUROC): its lead weights define Ground and Burst. The 2015-22 calculator runs the same four inputs, fit on that era.
DF_v15_leaderboard.csv	514 × 19	Published leaderboard: Steal+, Burst, percentiles.
DF_v15_reliability.csv	8 × 2	How much of one season's Steal+ is signal vs chance.
DF_v15_reliability_top.csv	8 × 6	Top runner-seasons with chance bands.
DF_v15_validation.csv	21 × 4	Validation behind every season-metric claim.
v15_players.json	—	The site payload (league fits, validation, players), synced into docs/index.html.

4. Figures

Figure	Used in
2-Data-Analysis/figures/1-Paper-GLMM	
Fig_paper_anatomy.png	paper, Figure 1; notebook
Fig_paper_breakeven.png	paper, Figure 2; notebook
Fig_paper_decision_value.png	paper, Figure 10; notebook
Fig_paper_drivers.png	paper, Figure 3; notebook; README
Fig_paper_greenlight.png	paper, Figure 11; notebook
Fig_paper_ground.png	paper, Figure 6; notebook
Fig_paper_influence.png	paper, Figure 5; notebook
Fig_paper_prediction.png	paper, Figure 4; notebook
Fig_paper_runners.png	paper, Figure 8; notebook
Fig_paper_speed.png	paper, Figure 7; notebook
Fig_paper_variance.png	paper, Figure 9; notebook
2-Data-Analysis/figures/2-v16-Logistic	
Fig_v16_tuning.png	notebook; technical report (archived)
Fig_v16_weights.png	notebook; technical report (archived)
2-Data-Analysis/figures/3-Engines-v12-v13	
Fig_v12_AUC.png	notebook
Fig_v12_ROC.png	—
Fig_v13_Attempt.png	notebook; technical report (archived)
2-Data-Analysis/figures/4-Studies	
Fig_add_delivery_beta.png	notebook; technical report (archived)
Fig_add_jump_decomp.png	technical report (archived)
Fig_add_jump_runners.png	technical report (archived)
Fig_add_jump_tests.png	notebook; technical report (archived)
Fig_add_needle.png	notebook; technical report (archived)
Fig_add_pitchtype.png	—

Figure	Used in
Fig_add_prior_delivery.png	notebook; technical report (archived)
Fig_add_speed_suppression.png	notebook; technical report (archived)
2-Data-Analysis/figures/5-Public-Baselines	
Fig_cmp_baselines.png	notebook; technical report (archived)
Fig_cmp_ladder.png	notebook; technical report (archived)
Fig_cmp_league_rates.png	notebook; technical report (archived)
Fig_cmp_mediation.png	technical report (archived)
Fig_cmp_runner_level.png	technical report (archived)
Fig_cmp_scenarios.png	technical report (archived)
Fig_cmp_separation.png	technical report (archived)
Fig_cmp_speed_by_season.png	notebook; technical report (archived)
Fig_cmp_speed_vs_ground.png	notebook; technical report (archived)
Fig_cmp_timeline.png	notebook; technical report (archived)
5-Live-Webapp/figures	
Fig_AUC.png	—
Fig_Importance.png	—
Fig_Metric_Cards.png	—
Fig_ROC_Calculator.png	notebook
Fig_v15_Decomposition.png	—
Fig_v15_Evidence.png	—
Fig_v15_Reliability.png	notebook; technical report (archived)